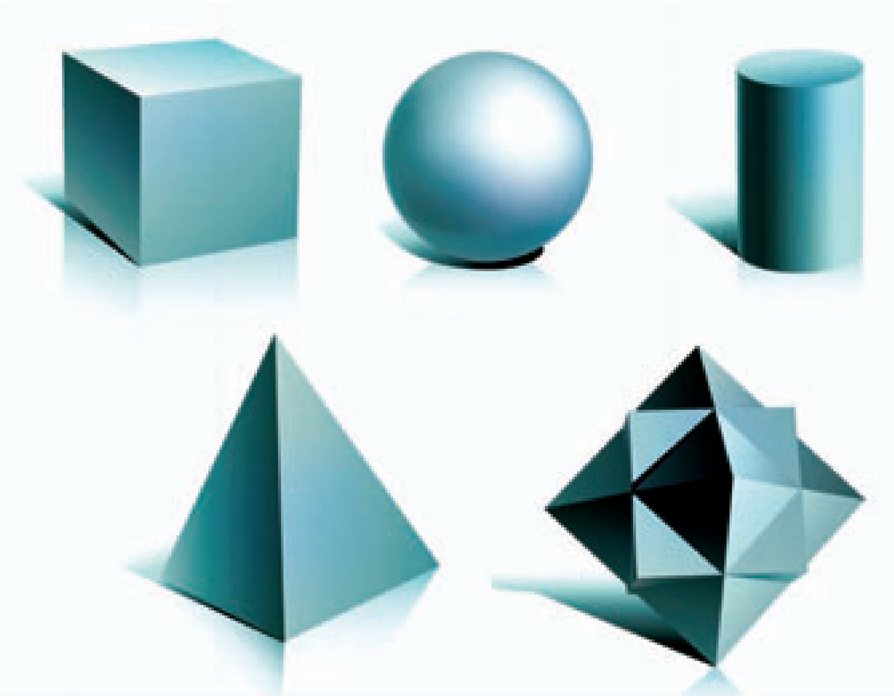

In the depth of the matter....

Everything you wanted to know about 3D printers but were too busy to ask

Madhusudan Mukerjee
readersletters@thinkdigit.com

Blame it on that German genius, Gutenberg. Around 1440, Johannes Gutenberg created the movable type machine and changed the world, exploding into a googol effect that Marshall McLuhan, the popular media theorist, called the 'Gutenberg Galaxy'.

From the first press, printing technology has gone through several changes, including the rotary press, the screen printer and the dot-matrix to culminate in today's compact and efficient inkjet and laser printers. But if you thought that this was the end of the line it's time for a serious rethink. Tomorrows printers don't just print text and images. They give you the real thing – the thing itself. To further use a convenient cliché, when it comes to 21st Century technology, tomorrow is today.

This 1954 German stamp shows Gutenberg at his printing press

RP Machines

'Rapid Prototyping' is an umbrella term that refers to a category of technologies that automatically build actual physical 'prototypes' – or models based on the data provided by a CAD (Computer Aided Design) software. 3D printing is one such technology. Engineers and designers have always needed prototypes to check out whether their concepts are viable in the real world or not. Most of these technologies were 'subtractive' – 'Machining', for example, is a technology wherein a nondescript block of material (typically, of metal or wood) is taken and the unnecessary parts removed (subtracted) to attain the required form. This is a really long process and could take several days or months, depending on the size of your prototype.

Rapid Prototyping, on the other hand, works somewhat in the reverse manner – an 'additive' process (but don't take the word 'rapid' too seriously). All RP machines use the same working principles: basically, the machine receives parameters of the object from a computer generated image and, splitting it into layers, proceeds to create these layers bottom first, adding one upon the other, until the object is formed. Instead of cutting, they heap. In short, it's a five stage process:

1. Make a virtual model using CAD software - anything from SolidWorks to SketchUp would suffice.
2. Convert it into an STL format. Since every CAD package uses different algorithms, the STL (stereolithography) file format has been accepted as the standard for all 3D printers universally to maintain consistency. Exporting the design file to STL transforms the object's image into a triangulated network so that the flat surfaces are represented in 3D. This conversion entails the introduction of the Z-axis. The greater the Z-value, the longer the production process. The same goes for the number of surfaces – and curved surfaces mean a lot of triangles, and thus, heavier files.
3. Use the proprietary software (usually provided by the manufacturers of the 'printer') to slice the model up into layers of minimal thickness. This allows you to control the build orientation keeping in mind the smallest detail. The program may also be used to set up support structures during the building process, allowing you to make partitions, nooks and crannies.
4. Construct the model using a chosen material. Typically, the machines use polymers (liquid or powdered), paper, or powdered metal. Almost any metal (Cobalt Chromium, Stainless Steel, etc.) would do. Photopolymers based on